

Aditya Kumar Singh

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Professional Summary

Linux Kernel & Embedded Systems Engineer with 5+ years of experience across Amazon and Qualcomm. At present, SDE-2 at Amazon working on first-party silicon platforms and ARM-based firmware environments. Former Qualcomm engineer specializing in IEEE 802.11 driver development (ath11k/ath12k) with upstream Linux kernel contributions and strong expertise in embedded Linux systems.

Education

Indian Institute of Technology Madras

MASTER'S DEGREE IN COMPUTER SCIENCE AND ENGINEERING (SPL. IN INFORMATION SECURITY, WEB ENABLED)

- Secured a CGPA of 8.81 out of 10

Chennai, Tamil Nadu, India

Aug 2023–Jul 2026

National Institute of Technology Tiruchirappalli

BACHELOR'S DEGREE IN ELECTRICAL AND ELECTRONICS ENGINEERING

- Secured a CGPA of 8.85 out of 10

Tiruchirappalli, Tamil Nadu, India

Jul 2017–May 2021

Kendriya Vidyalaya Tatanagar (CBSE affiliated)

HIGHER SECONDARY EDUCATION

- Secured 92.6%

Jamshedpur, Jharkhand, India

Apr 2016–Mar 2017

Kendriya Vidyalaya Tatanagar (CBSE affiliated)

SECONDARY EDUCATION

- Secured a CGPA of 10 out of 10 (95%) in Tenth grade

Jamshedpur, Jharkhand, India

Apr 2015–Mar 2016

Achievements

- Received **Breakthrough Award** at Qualcomm (Apr 2025). The award recognizes employee's contributions to product innovation and cost saving initiatives.
- Received **Emerging Star Award** at Qualcomm (Oct 2023) and (Oct 2024). The award recognizes the employee who consistently go above and beyond their work and demonstrate reliability, perseverance, and focus on results.
- Received **Rockstar Rookie Award** at Qualcomm (Jan 2023). The award recognizes the best hires from college graduates for their exemplary performance and who has significantly contributed to Qualcomm's success in very short span of time and embraced the Qualcomm values.
- Secured **1st place** in Elecart and SCDC events at **IIT-M Shaastra'20**, a National Level Techno-management fest.
- Secured **4th rank** in **eYRC-2018**, a National Level Robotics Competition organized by **e-Yantra, IIT Bombay**.
- Finalist** of **Sangam**, an intra-college **Hardware Hackathon in Pragyani'19** (The International Techno-Management Festival of NIT-Trichy).

Work Experience

Software Development Engineer (SDE) II

Amazon Development Center India Pvt. Ltd

Kernel Developer

Dec 2025 – Present

- Working as core member of BSP team for Amazon's first-party silicon chips powering Echo, Fire TV, and Kindle devices.
- Owning pin control, GPIO, and I2C subsystems, enabling board bring-up and peripheral integration.
- Gained hands-on exposure to ARM boot firmware stack, including Trusted Firmware-A (ATF) and SCP firmware, understanding boot sequence, power management initialization, and secure monitor interactions.

Senior Software Engineer

Qualcomm India Pvt. Ltd.

Linux Wireless Driver Developer

Jun. 2021 – Nov 2025

- Worked on latest IEEE 802.11 wireless standards such as 11be (Wi-Fi 7), 11ax (Wi-Fi-6), and contributed to Ath12k and Ath11k wireless drivers development.
- Alongside driver, also contributed to user space tools such as Hostapd, Supplicant, iw.
- Have gained knowledge on Regulatory, MLME, Data Path.
- Worked on enabling Multi-Link Operation (MLO) on upstream Ath12k.
- Implemented a critical feature of adding Wi-Fi 6E support to Ath11k driver.
- Worked with customers like Google, Eero (Amazon) for their launches which has ath12k/ath11k drivers.
- Open-Source Contributions:**
 - Linux Wireless Patchwork: [Link 1](#) | [Link 2](#)
 - Git repo: [ath](#) | [hostapd](#) | [iw](#)

Summer Internship at Hewlett Packard Enterprise

HPE

Vega

May, 2020 – Jul, 2020

- The project aimed to develop an AI-driven management system for primary storage. It was an ongoing live project hence a lot of new technologies were being tested to suit Vega's goal. Worked on testing the performance of technologies such as Kafka, Redis, Elasticsearch, etc. and wrote codes interfacing these using their Go-lang clients and getting the performance output.
- NOTE: Due to the COVID-19 pandemic, the internship was a work from home internship.

Summer Internship under Prof. Kavi Arya at e-Yantra, IIT Bombay

e-Yantra

Bio-Inspired Sub—Carangiform FishBOT

May, 2019 – Jul, 2019

- The project aims to design, develop, and build a bionic sub-carangiform fish robot that can swim effectively and is controlled through a remote. The robot has all basic features of a fish i.e. to go forward, take turns with variable radius and speed. The robot is cost-effective and has a battery backup of about 30 minutes.
- **GitHub Link:** <https://github.com/adityasingh3007/Bio-Inspired-Sub-Carangiform-FishBOT>

Projects

Automation system for easy upstreaming of patches

Qualcomm

Python, Shell

Jul, 2024 – Present

- Architected and actively maintaining a robust build automation framework inspired by OpenWrt, designed to generate customized Linux-based images from vanilla upstream repositories. The system enables seamless integration of private patches, supports extensive kernel and firmware customization, and streamlines the deployment and validation of images on embedded target devices. This framework significantly reduces manual effort, ensures reproducibility, and accelerates development cycles for faster upstreaming of patches.

E-mail bot to crawl internal mailing list

Qualcomm

Python, Shell

Jan, 2025 – Present

- Designed and developed an internal automation bot to streamline the validation and review process for wireless kernel patches. The bot monitors mailing lists and patch submission platforms, automatically applies patches to appropriate kernel trees, and triggers an automated validation pipeline. This includes code style checks (checkpatch), compiler validations (e.g., cross-compilation, 32-bit builds, allmodconfig), and static analysis. Test results are posted back directly as replies to the original patch emails.

AntBOT

e-YRC - 2018

Image Processing, 3D Designing, PID Control, Path Planning Algorithms

Nov, 2018 – Mar, 2019

- To understand the cooperation and coordination in the work performed by the ants, AntBOT aims to make a BOT which will depict the work of a single ant. The BOT will collect leaves, honey, wood from the Shrubs Area and store it for winter and/or remove the trash from its Ant Hills. The BOT will follow a pre-defined black path (same as path left by ants while traveling to collect supplies so that their fellow ants can follow it).
- **GitHub Link:** <https://github.com/adityasingh3007/AntBOT>

Skills

Programming Languages

C, C++, Embedded C, Python, Shell Script, JavaScript

Other Tools and Technologies

Git, Gerrit, OpenWrt, Buildroot, Yocto

Languages Know

English, Hindi

Position of Responsibility

President, Spider R&D Club of NIT-Trichy:

Apr, 2020 – May 2021

Member since Jul 2018

Have participated in the projects/activities related to Embedded Systems, Robotics and Control, Signal Processing, and Machine Learning through various national and college-level competitions.

Workshop Coordinator, Currents:

Aug, 2018 – Apr, 2021

Currents is an annual Electrical and Electronics Engineering department's symposium NIT-Trichy. Organized and conducted various workshops during Currents'19.

Extracurricular Activity

- Won 1st Prize in Carrom in Sportsfete-18, an inter-department sports fest of college.
- Part of the NSO team of NIT Trichy.